Problem 4. [1] Use the change of variable formula to show that for any $a, b > 0$, we have

$$\int_1^a x^{-1}\mathrm{d}x = \int_b^{ab} x^{-1}\mathrm{d}x.$$

Use this to argue that for $a, b > 0$

$$\int_1^{ab} x^{-1}\mathrm{d}x = \int_1^a x^{-1}\mathrm{d}x + \int_1^b x^{-1}\mathrm{d}x.$$

Solution. For the first equation, we use the substitution $u = bx$. Then $x^{-1} = (u/b)^{-1} = b/u = b \cdot u^{-1}$, and $b\mathrm{d}x = \mathrm{d}u$, the expression $x^{-1}\mathrm{d}x$ is equivalent to $u\mathrm{d}u$. We multiply the bounds of integration by b, obtaining

$$\int_1^a x^{-1}\mathrm{d}x = \int_b^{ab} u^{-1}\mathrm{d}u$$

We can always rename the variable of integration, so this is equivalent to

$$\int_b^{ab} x^{-1}\mathrm{d}x$$

as required.

Here is an alternative way to look at this: consider the function $f(x) = bx$. Then

$$x^{-1} = \frac{b}{f(x)} = \frac{f'(x)}{f(x)}$$

Using the change of variable formula, we can rewrite this as follow:

$$\int_1^a x^{-1}\mathrm{d}x = \int_1^a \frac{f'(x)}{f(x)}\mathrm{d}x = \int_{f(1)}^{f(a)} \frac{1}{u}\mathrm{d}u = \int_b^{ab} x^{-1}\mathrm{d}x$$

For the second equation, we first have

$$\int_1^{ab} x^{-1}\mathrm{d}x = \int_1^b x^{-1}\mathrm{d}x + \int_b^{ab} x^{-1}\mathrm{d}x$$

Using the first equation, this is equivalent to the desired sum of integrals:

$$\int_1^b x^{-1}\mathrm{d}x + \int_1^a x^{-1}\mathrm{d}x$$

[1] In this problem, you should not appeal to any properties of the logarithm function. In fact, you should not use an antiderivative of x^{-1} at all.